MILITARY SPECIFICATION
SYNCHRO CONVERTER SET, A-NEW

THIS SPECIFICATION HAS BEEN APPROVED
BY THE
BUREAU OF NAVAL WEAPONS, DEPARTMENT OF THE NAVY

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be designed to convert angular data represented by three wire 400 cycle per second (cps) electrical synchro signals into 11 bit binary digital data and to convert 11 bit binary digital data into three wire 400 cps synchro electrical signals.

1.2 Classification - The Synchro Converter Set shall consist of the items listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.10 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Microelectronic items shall conform to WR- 101, part 2 and the requirements specified herein.

(2) Other parts and materials requirements shall conform to Specification MIL-E-5400.

(3) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(4) Digital input amplifiers shall be designed in accordance with the requirements of Appendix A.

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3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.2.2 Electron Devices - Transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.2.3 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall be described by a contractor procurement specification and must be approved by the procuring agency. Microelectronic Modular Assemblies (Microelectronic Circuits) (MICS) shall be contained within Maintenance Modules, (for example, or printed circuit cards). The choice of Microelectronic Modular Assemblies shall be based on at least two sources of supply, which can be used interchangeably.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 50 pounds.

3.3.2 Reliability

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a total operating life of 50,000 hours with reasonable servicing and replacement of parts. Operating life is defined as that period of time during which the equipment has an MTBF equal to or greater than the minimum acceptable MTBF. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability Requirements

3.3.2.3.1 Design Mean Time Between Failures - The specified or design mean-time-between-failures (MTBF) shall meet or exceed 4,500 hours. The speciifed MTBF shall be substantiated by an analytical model of equipment reliability. The reliability model shall be prepared in accordance with the procedures established in MIL-STD-756 and MIL-HDBK-217. Failure rates and failure rate curves used shall be substantiated by minimum of 10×10^6 component hours for each component type or shall be taken from the appropriate curve in MIL-HDBK-217. The reliability calculations with the substantiating failure rate data shall be submitted with the contractors proposal to the government.

3.3.2.3.2 MTBF To Be Demonstrated By Test - For purposes of demonstrating MTBF by test, as opposed to analytical substantiation, 1500 hours shall be used as the specified MTBF when testing in accordance with paragraph 4.5.

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3.3.2.4 Materials and Process Reliability - All materials, components and processes shall be qualified to the maximum environmental conditions of MIL-E-5400, for Class 1 equipment. Qualification of materials, components and processes shall be conducted prior to their use in deliverable equipment. Qualification records and test results shall be maintained and made available to the Government upon request.

3.3.2.5 Parts Specifications - The contractor shall control the procurement of all parts and materials by detail contractor specifications. The specifications shall contain sufficient information and quality assurance requirements to preclude contractor acceptance of materials not in conformance with the equipment design and environmental requirements.

3.3.3 Maintainability - To provide for increased system effectiveness, maintainability shall receive equal consideration with such factors as cost, physical parameters, and reliability. The Maintenance Philosophy for avionics equipment procured for use in the A-NEW program is that of fault isolation to and repair by replacement of individual modules at the organizational maintenance level without removing equipment from the operational environment in the aircraft. Where feasible, fault isolation shall be computer initiated utilizing a diagnostic program; where not feasible, equipments shall provide self test features or readily accessible test points to accommodate standard test equipment affording fault isolation to the individual replaceable module. It shall be possible in 95% of the cases to fault isolate, repair and check-out the unit within a maximum of 30 minutes at the organizational maintenance level. In addition, the capability for performance testing shall be built into each system, subsystem and equipment to provide preflight readiness indications and to check performance levels when an out-of-tolerance condition is suspected.

3.3.3.1 DEFINITIONS

3.3.3.1.2 Terms used in this specification are defined in MIL-STD-280 unless herein noted.

3.3.3.1.3 Maintenance Levels - Maintenance levels shall be as defined in BUWEPS Instruction 4700.2a.

3.3.3.1.4 Module - The least complex segment of a component (Unit, Black Box) that can be removed and reinserted into the component through the use of connectors without a requirement for soldering (brazing) or welding. It is the assembly to which a fault is isolated in the aircraft. Modules may be either throw-away or shop repairable.

3.3.3.2 Requirements

3.3.2.1 Readiness Measurement Requirements - Suitable features shall be incorporated in the equipment to provide the operator, while airborne and while on the ground, with GO NO-GO indication of equipment readiness. These features may be in conjunction with, but not necessarily as part of the fault isolation features.

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3.3.3.2.1.1 Mode of Operation - The operation of the performance/readiness test feature shall be accomplished automatically and continuously by the computer unless it can be demonstrated that an on-demand, manually energized test mode of operation is more feasible.

3.3.3.2.2 Maintainability Requirements

3.3.2.2.1 Operational Maintainability Requirements - The time to repair shall be such that it will be possible in 95% of all the cases of failures, to perform corrective organizational maintenance actions within 30 minutes. As here applied on organizational corrective action includes the following:

(1) Recognition of a fault

(2) Isolation of the fault to a module

(3) Module replacement

(4) Check out of the equipment

3.3.3.2.2.2 Packaging Requirements - The equipment shall be packaged in modular form so that, to the maximum extent practical, all sub-assemblies shall be of the plug in type and should be easily removable without special tools, for maintenance, If a special tool is required for removal, it shall be provided and attached to the equipment.

3.3.3.2.2.3 Module Commonality - The number of different modules will be kept to an absolute minimum. This standardization is mandatory to ease the testing and spares requirements at all levels of maintenance.

3.3.3.2.2.4 Module Accessibility - Module arrangement, will be such that access to each module does not require the removal of adjacent modules or parts other than access panels.

3.3.3.2.2.5 Keying - Modules and connectors shall be keyed to prevent the insertion of a connector or module into an improper location within or on the equipment.

3.3.3.2.2.6 Part Replacement - Part replacement on the analog modules shall be possible by skilled technicians working in a designated service area such as intermediate maintenance activity. Use of potting compounds, excessive moisture/fungus proofing compounds or any other substance to mechanically support piece parts is forbidden. Moisture/fungus proofing compounds, not in excess of a 2 mil thickness is permitted.

3.3.3.2.2.7 Non-repairable assemblies - All designs for non-repairable assemblies shall be submitted to the Government, for approval, prior to release for fabrication.

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3.3.3.2.2.8 Adjustments - In ord r to decrease the complexity of the maintenance tasks the equipment should be designed to require no periodic adjustments, alignment or calibration. Adjustments shall be capable of being made in the aircraft using simple hand tools and/or General and Standard Test Equipment. Adjustments shall be identified and accessible without a requirement to remove or relocate the module, any other modular or part.

3.3.3.2.2.9 Access - It shall be possible to perform 90% of the anticipated maintenance actions by access to one side of the equipment only.

3.3.3.2.2.10 Access Panels - Access panels shall be retained by quick disconnect fastners i.e. hand operated slide, dzus type etc.

3.3.3.2.2.11 Air Filters - Devices requiring removal periodically for inspection and/or servicing shall be retained by quick disconnect fastners i.e. hand operated slide, dzus type etc.

3.3.3.2.212 Mounting Devices - Connectors, receptacles, fuses, circuit breakers or any other electronic part, shall not be a part of any mounting device associated with the equipment.

3.3.3.2.2.13 Aircraft Connectors - Aircraft cables shall mount directly to the equipment, using quick disconnect type connectors.

3.3.3.2.2.14 Aircraft Accessibility - Form factor and location of connectors, air filters, circuit breakers, access panels, and modules shall be such that when the equipment is installed in the aircraft, all in-aircraft maintenance actions may be performed with the equipment in its installed position.

3.3.3.2.2.15 Protective Devices - Circuit breakers shall be used in lieu of fuses whenever possible. Fuses, if used, shall be of types and ratings readily available from Navy Supply. Fuse holders and/or circuit breakers shall be accessible from the exterior of the equipment, plainly marked as to rating and size, and shall be of the fault indicating type and/or have access for quick voltage checks.

3.3.3.2.2.16 Handles - Handles shall be provided as part of the equipment. They shall be located to give optimum handling capability by one man.

3.3.3.2.3 Fault Isolation/Manual Preparation

3.3.3.2.3.1 Fault Isolation - In conjunction with, but not necessarily as part of the Performance/Readiness testing features, there shall be features incorporated which will permit fault isolation to the module level. These features shall include the necessary switching arrangements with a GO NO-GO readout device that will clearly indicate whether or not the function selected is within minimum performance standards. If a built-in readout device is not feasible, means shall be provided to use general or standard test equipment, i.e. Oscilloscope, VTVM etc.

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3.3.3.2.3.2 Diagnostic Ability - If feasible, the equipment shall be logically and electrically designed to facilitate the location of malfunctioning modules through diagnostic programming in conjunction with the aircraft's computer and/or other installed electronic equipment. Test points shall be made available at the front of the equipment to assist diagnosis of malfunctions.

3.3.3.2.3.2 Diagnostic and Maintenance Routines - The contractor shall assume responsibility for preparation and delivery of the diagnostic routines, and a program listing including the description of use for each as well as preparation and delivery of the associated maintenance manual.

3.3.3.2.3.3 Manual Preparation - The equipment maintenance manual to be used in the performance of the testing will be written to a comprehensive ability level as described in paragraph 3.7.6 of Naval Weapon Specification, WS 3099.

(1) Civilian Education - High School graduate or equivalent.

(2) Technical Training - Navy, Class A technician school and training courses appropriate to the maintenance task.

(3) Experience - Two years of technician experience in addition to technician training.

3.3.3.2.4 TEST POINTS

3.3.3.2.4.1 Test Point Requirement - All test points shall accommodate standard Navy approved test equipment, i.e. multimeter or oscilloscope probes. In the event miniature test points cannot accommodate standard probes, use of adapters will be permitted. These adapters are to be provided with and secured to the set.

3.3.3.2.4.2 Power Supply Test Points - Accessible external individual test receptacles shall be provided on the equipment, to measure the output voltages. The receptacles shall be identified as to the voltage normally present at the test point.

3.3.3.2.4.3 Fault Isolation Test Points - Test points are required to provide capabilities for testing, when nature of malfunction is beyond scope of computer diagnostic test program, built-in-test equipment (BITE) or when other provisions have not been made for fault isolation to the function and/or module level. In addition, for those equipments incorporating BITE, test points are required, in the event of BITE failure, to repair the BITE. The test points shall be accessible from the front of the equipment, and when individual test points are not feasible, these test points shall be grouped in multi-pin receptacle(s) A/F, M38300/3603-10, with caps provided. A diagram, of durable construction, showing waveforms, voltages and etc., as related to each pin, shall be attached to each unit.

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3.3.3.2.4.4 Shop Repair Test Points - Test points shall be provided on the equipment and/or module for use when the equipment or module is removed from its next higher assembly, or is newly issued, to determine the performance level, and in the event of failure, to determine which subassembly, circuit or piece part within the equipment or module is at fault. They should provide the means for:

(1) Measuring the necessary input/output and circuit parameters to enable isolation to the faulty subassembly or piece part.

(2) Measurement of necessary input/output parameters to enable calibration of alignment of the equipment or module.

3.3.3.2.5 Special Support Equipment - Self test features shall be such as to obviate the need for any Special Support Equipment for preflight; post flight checks or fault isolation. There shall be no special support equipment for support of the equipment except as expressly authorized by the government.

3.3.3.2.6 Time Totalizing Meter - The equipment shall contain a time totalising meter in accordance with Specification MIL-M-7793. The meter size shall be in keeping with the size of the equipment.

3.3.3.3 Maintainability Program

3.3.3.3.1 Maintenance Plan - Within 30 days subsequent to award of contract, the contractor shall supply the Government with the:

(1) Names of personnel of the organization which is responsible for satisfying the requirements of paragraph 3.3.3. of this document presented in sufficient detail to indicate relative position, functions, responsibility and authority.

(2) Plan for designing, developing, and integrating maintainability into the equipment and subsequent demonstration and verification of maintainability requirements.

(3) Packaging information pertinent to organization level maintenance action, such information is to include but not be limited to; form factors, method of mounting, weight and location of external connections, fuses, circuit breakers, test points, access panels and covers.

3.3.3.3.1.1 Planning Conference - At a date mutually acceptable to both the procuring agency and the contractor, but in any event not later than 20 days subsequent to submittal of the contractor's maintenance plan, the procuring agency's respresentatives shall meet with the contractors for the purpose of reviewing, modifying, and approving the contractor's plans for satisfying the requirements of this document.

3.3.3.3.2 Maintainability Assurance - Testing to demonstrate the achievement of the maintainability requirement specified herein shall be in accordance with 4.7.

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3.3.4 Cabling and Connections -

3.3.4.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400 and 3.5.3.9.

3.3.4.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment. Digital signals shall be transmitted in twisted pair type cables.

3.3.5 Interchangeablility of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference Control - The equipment shall conform with all of the requirements of WR 101, part 1 and MIL-I-6181D for design, construction and workmanship except for paragraphs 4.3.3, 4.3.4.4, and 4.3.4.5 of MIL-I-6181D. The contractor's interference control program shall include but not be limited to the development and submission of the following documents to the procuring activity for approval.

DOCUMENT	SUBMISSION DATE
Interference Control Plan	90 days after receipt of contract
Test Plan	60 days before scheduled start of pre-production test.
Test Report	30 days after completion of pre-production test.

3.3.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C±5°C)
Altitude	Normal ground
Vibration	None

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Humidity	Room ambient up to 90% relative humidity
Input power voltage	115± 1.0 V AC

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 A equipment, except as modified herein:

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400.

3.3.10 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be dept to a minimum and shall not exceed 0.5 minutes under standard conditions and 1.0 minute at extreme service conditions.

3.3.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power source with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

AC Power (three phase) 115/200V, Category "B", 125 VA

3.3.11.1 Input Voltage Transients - The input voltage transients shall remain within limits 1 and 3 of Figure 1 of MIL-STD-704, except that limit 3 shall not go below 80 volts. Transients extending below 80 volts for any time duration shall not cause equipment degradation or failure and, upon return of the input voltage within operating limits, the equipment shall automatically recover.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein.

3.4.1 Operation - The Synchro Converter Set shall be designed to convert any one of 32 three wire 26 volt excited synchro transmitter electrical signals into a single 11 bit binary electrical digital output signal representation of the selected channel selected by the computer. The output binary digital signal shall be suitable for transmission of data into the central digital computer. Similarly, the Synchro Converter Set shall be designed to convert, as commanded by the central digital computer, 11 bit digital signals transmitted to the Synchro Converter Set into three wire 26 volt excited synchro transmitter electrical signals. The signals shall be suitable for use by any one of 16 three wire 26 volt excited synchro receiving units. The interrelationship of input and output signals is shown in Figure 1.

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3.4.2 Number and Type of Analog Channels - The Synchro Converter Set shall be capable of accepting signals from 32 three wire 26 volt excited synchro transmitters and transmit signals to 16 three wire 26 volt excited synchro receiving units. The above input and output channels shall be modular and interchangeable without adjustment, between similar types.

3.4.3 Angular Binary Representation and Accuracy of Conversion - The synchro angular position shall be represented by an eleven bit parallel true binary word. The angular range shall be 0° to 360° with 0° = 0000 (octal) and 359° 49.5' = 3777 (octal). The accuracy of converting an angle represented by an 11 binary input digital signal into a three wire output synchro signal shall be such that the converted output signal corresponds to the input angular data within ± the least significant bit of the 11 bit output binary digital signal.

3.4.4 Insensitivity to Voltage Variations - The performance of the Synchro Converter Set shall not be affected by variations either in the 400 cycle excitation voltage, frequency, and harmonic content or in 400 cycle three wire input signal voltage, frequency, and harmonic content within the limits as specified in MIL-STD-704 Category B for normal operation, except as modified in para. 3.3.11.1.

3.4.5 Intercommunication Signals - The intercommunication signals between the Synchro Converter Set and the A-NEW central digital computer as illustrated in Figure 1 shall consist of control and request signals and parallel data lines. The intercommunication signals shall consist of output and input request signals between the Synchro Converter Set and the computer output and input acknowledge signals as well as input/output data lines between the computer and the Synchro Converter Set. The intercommunication signals shall be as specified in specification MIL-C-81332(WP) for the A-NEW digital computer, and as further described in sections 3.5.1.2 and 3.5.2.1 herein.

3.4.6 Form Factor - The Synchro Converter Set shall be enclosed in a single case which has been designed in accordance with specification MIL-C-172. The dimensions of the case shall not exceed the following:

Length - 15 inches

Width - 15 inches

Height - 10 inches

However, the Synchro Converter Set may be smaller than the above.

3.5 Detail Requirements - The Synchro Converter Set shall be furnished as a single unit and contain power supplies, conversion units, and such other circuit assemblies as required to produce the two principal functions of synchro to digital conversion and digital to synchro conversion as described

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synchro to digital conversion and digital to synchro conversion as described in section 3.5.1 and 3.5.2 respectively. The Synchro Converter Set shall be designed to operate as an intermediate unit between synchro transmitter and receiver units as specified in specification MIL-S-20708 and the A-NEW digital computer.

3.5.1 Synchro to Digital Mode - In the synchro to digital (S/D) mode the conversion equipment shall be designed to receive three wire 400 cps electrical signals simultaneously from 32 synchro transmitters of 26 volts excitation. The Synchro Converter Set shall be designed to receive instructions from the computer as to which of the 32 synchro channels is to be converted into binary form and then convert the data on the synchro channel specified into 11 bit binary form for transmission to the computer.

3.5.1.1 Synchro Input Signal Characteristics - The Synchro Converter Set shall be designed to receive simultaneously 32 synchro transmitter three wire synchro 400 cops electrical signals. The three wire signals may vary from 0 to 11.8 volts as received from 26 volt excited synchro control transmitters as described in section 3.4.4 and specification MIL-S-20708. The loading presented by the Synchro Converter Set line to line on each pair of the three wire synchro inputs shall be 5000 ohms minimum. The loads on the three paired wires for each three wire synchro input shall be balanced to each other within ± 2%. If the impedence is greater than 5000 ohms. The Synchro Converter Set shall be capable of accepting and converting synchro signals at the required accuracy (see 3.4.3) where the modulation of the 400 cps synchro signal varies at a rate caused by input synchro control transmitter rotational speeds from 0 to 20 revolutions per minute. The 26 V excitation will be an unregulated voltage derived from 115 V specified in paragraph 3.3.11.

3.5.1.2 Control and Data Request Signals - The Synchro Converter Set, in the S/D mode shall be designed to accept computer control signals and to provide data request signals in accordance with specification MIL-C-81332(WP) in order to permit the following sequence of operations.

(1) The Synchro Converter Set raises the Output Data Request line to request channel selection data.

(2) The computer sets the channel address word in accordance with the code and format in Figure 2.

(3) The computer sends an Output Acknowledge signal to the Synchro converter Set to indicate the channel address is available.

(4) The Synchro Converter Set lowers its Output Data Request and External Function Request, see 3.5.2.1, and processes the channel specified.

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(5) When the S/D conversion is completed, the Synchro Converter Set sets the input data lines to the computer giving the synchro angular portion in accordance with the code and format shown in Figure 3 and advises the computer the converted data is ready for input by raising the Input Data Request.

(6) The computer reads the input data and sends an Input Acknowledge signal to the Synchro Converter Set.

(7) The Synchro Converter Set on receipt of the Input Acknowledge signal shall request a new channel address from the computer as in (1), raises the External Function Request, and the cycle repeats.

3.5.1.3 Rotor Signal - On each connector of the Synchro Converter Sets that receives information from a synchro transmitter, the Synchro Converter Set shall make available a 26 V 400 Cycle signal suitable for the rotor windings of the synchro. This signal will be available but not mandatory for use by the peripheral unit.

3.5.1.4 Conversion Time - The S/D equipment shall be designed to complete the conversion of synchro signals to 11 bit binary data suitable for transmission to the computer in 100 microseconds or less measured from the receipt of an Output Acknowledge signal per (4) of section 3.5.1.2 to the initiation of an Output Data Request for channel selection per (7) of section 3.5.1.2.

3.5.2 Digital to Synchro Mode - The digital to synchro (D/S) conversion equipment shall be designed to receive 11 bit binary angular data and 4 bit binary channel selection data from the computer and convert the data into 26 volt synchro control transmitter type three wire output signals for any one of 16 three wire output channels. The D/S equipment shall further be designed to maintain converted synchro signals on all 16 output channels to provide for simultaneous parallel excitation of 16 synchro receiving units.

3.5.2.1 Control and Data Signals - The D/S equipment shall be designed to accept computer control and data signals in accordance with specification MIL-C-81332(WP) in order to permit the following sequence of operations.

(1) The D/S equipment requests data from the computer by raising the External Function Request.

(2) The computer places 11 bit angular data and 4 bit channel selection data on its output data lines in accordance with the format and code shwon in Figure 4.

(3) The computer sends an External Function signal to the Synchro Converter Set advising that the requested data can be read.

(4) The Synchro Converter Set reads the data and drops its External Function and Output Data Request, see paragraph 3.5.1.2, signals to the computer.

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(5) The Synchro Converter Set converts the 11 bit binary data into three wire synchro signals for the channel selected by the 4 bit channel data.

(6) The Synchro Converter Set requests data from the computer as in (1) raises the Output Data Request and the cycle repeats.

3.5.2.2 Synchro Output Signal Characteristics - The Synchro Converter Set shall be designed to transmit simultaneously 16 synchro transmitter type 26 V excited 3 wire 400 cps electrical signals suitable for driving 16, 26 volt type synchro receiving units. The quality of the signals shall be equal to or better than the limits listed in section 3.4.4 for three wire synchro input signals to the Synchro Converter Set. The transmitter type output signals also shall be in accordance with the requirements of specification MIL-S-20708 for 26 volt 400 cps excited control transmitter three wire synchro output signals. Each of the 16 output channels of the D/S equipment shall be capable of driving one synchro receiving unit of the control transformer type, specified in specification MIL-S-20708. The impedance of the receiving synchro units may range in value from 100 to 600 ohms. The zero values and markings of the output synchro type three wire signals and terminals shall be in accordance with the provisions of specification MIL-S-20708 for 26 volt excited synchro control transmitters.

3.5.2.3 Conversion Time - The D/S equipment shall be designed to complete the conversion of 11 bit binary data into output three wire synchro signals in 30 microseconds or less as measured from the receipt of the signal that the requested 11 bit data is ready per (3) of section 3.5.2.1 and the raising of the External Function Request Signal.

3.5.2.4 Updating Rate - The D/S equipment shall be designed to provide continuous output three wire synchro type signals on each of 16 output channels.

3.5.3 Additional Requirements - Requirements in addition to those specifically stated in sections 3.5.1 and 3.5.2 for the Synchro Converter Set shall be as follows.

3.5.3.1 Cable Lengths - Provisions shall be made in the Synchro Converter Set for developing signals capable of being received at the peripheral equipment indicated in Figure 1 when this equipment is attached to cables from 1 foot to as long as 150 feet.

3.5.3.2 Reverse Signal Blocking - Means shall be provided in the digital input and output circuits of the Synchro Converter Set to block digital signals having signs reversed from those specified in specification MIL-E-81332 (WP) for the A-NEW Avionics Computer.

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3.5.3.3 Effect of Equipment Failure - The design of the equipment shall be such that a failure shall have no disruptive effect on the operation of the peripheral equipment in so far as the peripheral equipment can operate on other modes which do not require use of the Synchro Converter Set.

3.5.3.4 Effect of an Open Line - Digital input circuits shall be provided such that if an input wire is disconnected, the effect will be as though a logical "zero" were present at the input. Digital and Synchro output circuits shall be designed to be capable of driving into an open line without harmful effects.

3.5.3.5 Effect of a Shorted Line - Digital input circuits shall be provided such that a shorted or grounded line appears as a logical "one". Synchro and digital output circuits shall be designed to be capable of driving into a direct short circuit without exceeding the dissipation limits of any component.

3.5.3.6 Effect of Power Shut-Down - Digital Output circuits shall be designed so that they are effectively disconnected from the line when power is cut off due to deliberate or accidental power shut-down.

3.5.3.7 Microelectronic Circuit Design - All electronic circuitry in the Synchro Converter Set shall be fabricated in microelectronic form such as thin film, bulk semiconductor or a combination of both. Separately attached elements such as conventional transistors, diodes, capacitors, resistors, inductors or transformers shall not be used except as absolutely necessary due to limitations in the state-of-the-art of microelectronic techniques.

3.5.3.8 Ground Connections - A 3/8" stud fastened to the cabinet in a manner such as welding or brazing which results in a connection of very low resistance shall be provided for grounding the cabinets to the hull. All input and output lines shall be isolated from the cabinet ground by at least 1 megohm. Power ground shall be isolated from signal ground. Isolation transformers shall be provided for the synchro inputs and outputs.

3.5.3.9 Electrical Connectors - External - The following electrical connectors shall be provided.

(1) A connector conforming to MIL-C-38300/3654-10 shall be provided for cables that interface with the computer.

(2) A connector conforming to MIL-C-26482 shall be provided for all other circuits.

3.5.3.9.1 Channel Connectors - Each synchro channel shall be on a separate connector thus requiring a total of 48 connectors for analog information.

3.5.4 Power Supply Requirements -

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3.5.4.1 Input Power - The power supply shall operate on an input voltage of 115 volts line to neutral, 200 volts line to line, three phase, four wire "Y" system, 400 cycles per second as detailed in MIL-STD-704, category B mode of operation and as modified herein.

3.5.4.2 Transient Protection - In the event that the power supply cannot filter out severe power transients, such as encountered during generator switch-over sensing circuits shall be provided that will indicate to the A-NEW Avionics Computer when out-of-tolerance voltages are present. (See Figure 3)

3.5.4.3 Protection - The Power supply shall be designed to insure against excessive output over-voltage which might own as a result of an internal component failure within the power supply circuitry.

3.5.5 Cooling - Internally generated heat shall be dissipated to the atmosphere without the use of forced air cooling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any inspections set forth in the test specification where such inspections are deemed necessary to assure supplies and services conform to the requirements of this specification and the contract or purchase order.

4.1.1 Classification of Tests - Items covered by the specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Reliability Tests

(5) Parts Tests

(6) Maintainability Tests

4.1.2 Tests and Test Procedures - A list of proposed tests, test procedures and test equipment shall be furnished by the contractor to the Government inspector for approval 40 working days prior to initiation of the tests outlined herein.

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4.1.3 Remedial Action Guarantee - Remedial action shall be performed by the contractor, including retrofit on all delivered units at his expense, for all failures and equipment deficiencies detected during any of the Quality Assurance tests.

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.7. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient temperatures.

4.2.2 Scope of Tests -Preproduction tests shall include, as a minimum, environmental tests in accordance with the procedures of Specification MIL-T-5422, interference tests as specified in 3.3.6 herein, reliability tests as specified in 4.5, maintainability tests as specified in 4.6, and performance tests as specified in 6.10.

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipment shall be delivered prior to approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the production sample shall be at the contractor's own risk.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Incorporation of changes in the design, construction, or performance shall only be permitted based on prior approval from the procuring activity. Each production equipment shall be capable of successfully passing the same tests as imposed on the last prior preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and, for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests (see 4.4.1). The preproduction sample shall not be selected for this test.

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4.3.1 Scope of Tests - The equipment may be subject to any and all tests specified in the approved test procedures to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements specified herein.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory data (i.e. interconnection diagrams, schematic diagrams and previous test data) necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - Upon completion of the initial production tests, the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the equipment the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of the material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of individual tests.

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, and operational adequacy. As a minimum, each equipment accepted shall have passed the following tests:

(1) Manufacturing Run In Test

(2) Examination of Product

(3) Operational Test

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4.4.1.1 Manufacturing Run In-Test - Each equipment shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Cycled between the maximum specified limits as outlined by paragraph 4.6.7 cycle phase 2.
Shock	Three shocks in each direction each of three mutually perpendicular axes. Shocks pulse half sine wave, peak value of 15G and duration of 11 milliseconds.
Vibration	Any selected frequency within the range of 20 to 30 cps (exluding resonant points) and a maximum amplitude of ±3 g's.

The mechanical shock test consisting of a total of 18 shocks as defined above shall be performed immediately prior to the beginning of the 6-hour operating period. The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes at the beginning of the operating period. The temperature shall be cycled, between the maximum limits specified by paragraph 4.5.7 cycle phase 2, allowing enough time at each extreme to permit temperature stabilization within the equipment. The equipment shall be operatedduring these periods for the purpose of detecting flaws and imperfect workmanship. Operation shall consist of a combination of the tests in 4.4.1.3.3. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. Operations within the specified limits of satisfactory performance is not necessarily required during the vibration period. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration and temperature cycling periods need not be repeated when it is certain the failure was not a result of these tests. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.1.2 Examination of Product - Each equipment offered for delivery shall under-go such examination and testing as will prove the workmanship and reveal the omission and errors of the production process.

4.4.1.3 Operational Tests - Each equipment shall be subjected, as a minimum, to the following operational tests. These tests shall be conducted at normal ambient temperature.

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4.4.1.3.1 Manual Operations Test - The equipment shall be subjected to a manual operations test which demonstrates the functional ability of all controls and indicators to effect the equipment in the required manner.

4.4.1.3.2 Power Input Test - Each equipment shall be subjected to a range of input voltages and frequenceis as specified herein.

(1) The input voltage shall be varied within the limits as specified in 3.5.4.1, and the input frequency varied between limits 2 and 3 of Figure 3 of MIL-STD-704 with no effect on the operating test program. The test program shall be a combination of those in 4.4.1.3.3.

(2) It shall also be demonstrated that input voltage variations greater than those specified in 3.5.4.1 shall cause an automatic shut down of the equipment and an automatic restart upon voltage reinitiation.

(3) It shall also be shown that normal operation can be resumed.

4.4.1.3.3 Endurance Run - Each equipment shall be subjected to an endurance run of not less than 8 hours, with no errors allowed during this period. This run shall consist of a series of functional tests, exercising all portions of the equipment in a manner at least as stringent as that required during its ultimate application. The run shall consist of the following test, which shall each be exercised for a period of not less than eight hours overall. This test shall exercise the entire Synchro Converter Set and the following shall be monitored.

(1) Accuracy on all channels.

(2) Ability to input and output

4.5 Reliability Assurance Test - Reliability assurance tests shall be conducted in accordance with Military Specification MIL-R-23094 except as specified herein.

4.5.1 Quantity of Equipments to Be Tested - Two equipments shall be used for this test. The test shall begin as soon as the first equipment assigned to this test is available. The second equipment assigned to this test shall be added to this test when it becomes available.

4.5.2 Test Procedure - Procedure one (1) of Military Specification MIL-R-23094 shall be used for this test.

4.5.3 Accept-Reject Criteria - Figure 15 of Military Specification MIL-R-23094 shall be used as the Accept-Reject criteria. For purposes of demonstrating MTBF by test as opposed to analytical substantiation, 1500 hours shall be used as the specified MTBF when using Figure 15.

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4.5.4 Test Level - For each increment of 1650 hours of equipment testing, the equipment shall be tested for a total of 340 hours at Test Level III, Table I, of MIL-R-23094 and tested for 1310 hours at Test Level I, Table I.

4.5.5 Length of Test - The equipment shall be tested until an accept or reject decision is made in accordance with 4.5.3. If no decision has been made within the test time specified, the contractor and the procuring agency shall confer on further testing procedures and criteria. If the equipment is accepted within the test time specified in 4.5.3, the equipment shall continue to be run until a total equipment operating time of 13000 hours has been achieved. The MTBF calculated for the total operating time is not for the purpose of acceptance or rejection but rather for the purpose of more closely determining the actual MTBF of the equipment. Test Level I, Table I, of MIL-R-23094 shall be used for the post-acceptance portion of the equipment operating time. For this post-acceptance portion of the test all requirements pertaining to corrective action, failure analysis, and reporting shall still be complied with.

4.5.6 Procedure Following a Reject Decision - Equipments that have been rejected under this test may be reworked by the contractor at his expense to correct defects, and resubmitted for test. The contractor and the procuring activity shall mutually agree upon the particulars for retesting prior to its initiation.

4.5.7 Burn-in (Debugging)Period - Each article shall be subjected to the following environmental test sequence prior to being placed in the endurance cycle. Failures encountered during the environmental sequence shall be recorded, evaluated and corrected. These failure shall not be counted in the computation of MTBF.

(1) Temperature Cycle

Cycle Phase	1	2	3
Temp. Change	+85°C to -62°C	-54°C to +55°C	55°C (no change)
Equipment	Off	Performance testing	Performance testing
Duration	As Required for Temp. Stabilizing	As Required for Temp. Stabilizing	3 hours

(2) Shock - Three shocks in each direction shall be applied along the three mutually perpendicular axes of the test item (18 shocks). The shock pulse shape shall be half sine and shall have a peak value of 15G and a duration of 11.0 milliseconds. The test item shall be operating during and after the test.

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(3) Vibration - Vibration of 40 ± 20 CPS (sinusoidal) at an acceleration level of ± 2.0G (peak) shall be applied for 15 minutes in each of three mutually perpendicular axes of the test item. The test item shall be operating during and after the test.

4.5.8 Preventive Maintenance - No preventive maintenance shall be performed during this test except when approved by the procuring agency.

4.5.9 Performance Characteristics to be Measured - The performance characteristics to be measured under this test, as well as the detailed test procedures and all reporting forms, shall be submitted by the contractor to the procuring agency for approval at least 60 days prior to initiation of test.

4.5.10 Reliability Measurement

4.5.10.1 Operating Time

4.5.10.1.1 Included Time - The time accumulated by each equipment will be monitored by an elapsed time meter and an appropriate test log. "Operation" describes the condition where the equipment start sequence has been initiated and ends with failure, turn-off or shutdown.

4.5.10.1.2 Excluded Time - Operating time accumulated during any government approved period of special testing will be excluded.

4.5.10.2 Failure Definition

4.5.10.2.1 Failure - The inability of an item to perform its required function within previously established limits.

4.5.10.2.2 Independent Failure - A part failure which is not related to the failures of associated parts.

4.5.10.2.3 Dependent Failure - A part failure which is caused by the malfunctioning of associated parts.

4.5.10.2.4 Any unapproved preventive maintenance action shall be considered prima facie evidence of a failure.

4.5.10.3 Data Acquisition

4.5.10.3.1 Operating Time Data - Elapsed operating time will be monitored and recorded in the test log. Time meter readings will be identified by the date span covered and will be made at selected regular intervals.

4.5.10.3.2 Failure Data - Failure data will originate in the form of a failure malfunction report and will record all item malfunctions and discrepancies, including the time meter reading at the time of the detection and a detailed history of the resulting maintenance and repair. A detailed failure analysis

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will be conducted of the failed part. The contractor shall establish a corrective action loop and document the elimination of the assignable cause of failure. Where practicable, testing shall establish the effectiveness of the proposed corrective action. Corrective actions shall be approved by the Government prior to implementation.

4.5.10.4 Reliability Computation - Computation of achieved reliability will be conducted at joint data evaluation meetings between the contractor and the government. Failure data will be examined and the results of corrective action will be presented.

4.5.10.4.1 Non-Chargeable Failures -

(1) Failures classified as "chargeable" may be excluded from the computation by mutual agreement between the government and the contractor when:

(a) The contractor analytically and/or by test establishes the effectiveness of the corrective action (or change) to the satisfaction of the government.

(b) The proposed corrective action or change is approved by the government.

(c) The corrective action or change is scheduled for introduction and/or retrofit by the contractor as approved by the government. Approval of any change by the government shall not protect any subsequent failure from classification.

4.5.11 Reliability Engineering Analysis - Sixty days prior to subjection of the equipment to reliability testing, the contractor shall submit, for approval, to the procuring agency a detailed analytical reliability analysis of the equipment. The analysis shall show, but not be limited to electrical and thermal stress on each part, failure rates of parts, safety margins for parts and environmental effects on parts.

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4.5.12 Reliability Reporting - The contractor shall submit a report to the procuring agency each month. This report shall present the current status of the reliability program and define reliability problems, if any, and propose solutions. Succeeding reports shall carry the entries of previous reports if there exists a previously unsolved reliability problem. The contractor shall provide access to all test data upon request. Five copies of this report shall be delivered to the procuring agency within 10 days following the end of each month.

4.5.13 Failure Reporting - All failures shall be reported to the procuring agency within five (5) days of occurrence. The failure report shall include as a minimum:

(1) Report Number

(2) Equipment (WRA) Identification

(3) Subassembly Identification

(4) Component or Part Identification

(5) Time on Failed Part

(6) Preliminary Failure Analysis.

4.5.14 Failure Analysis - The contractor at his expense shall conduct a planned program of failure analysis throughout the life of the contract. All failures related to the test program or to deliverable equipment shall be analyzed to the level required to assure corrective action and to prevent reoccurrence. Failed microelectronic functional devices shall be photographed by micro-techniques to provide a graphic record and a vehicle for corrective action communication. Failure analysis records shall be included in the monthly reliability report and contain the following information as a minimum requirement.

(1) Report Number

(2) Reporting Activity and Individual

(3) Equipment Identification

(4) Major Assembly Identification

(5) Subassembly Identification

(6) Component or Part Identification

(7) Date of Failure

(8) Time on Failed Part/Item

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(9) Name of Test/Operation

(10) Environment

(11) Description of Trouble

(12) Failure Cause/Analysis

(13) Disposition (Detail)

(14) Name of Failure Analysis Authority

(15) Corrective Action Required

(16) Failure Effect Analysis

4.5.15 Final Reliability Report - A final reliability report shall be prepared to summarize the final results of the reliability program. The report shall be submitted to the procuring activity within 30 days after completion of the reliability test and shall include but not be limited to the following:

(1) Final operational reliability of the equipment under operational conditions stated in the test procedures.

(2) Measures taken by the contractor to increase the reliability of the equipment and the improvement in reliability attained by the measures taken.

(3) Conditions to be avoided if a high reliability factor is to be maintained in field operations.

(4) Quality control procedures necessary to maintain a high standard of reliability for equipment produced in quantity.

(5) A summary of all failure data and the effectiveness thereof.

4.5.16 Quantity of Reports - All reports required under this test shall be submitted in quantity of five each with one reproducible master.

4.6 Maintainability Assurance - The following procedure shall be used to demonstrate that the time-to-repair this equipment is not greater than 30 minutes for any single repair.

4.6.1 Test Conditions - One (1) equipment shall be used for this test. The contractor shall select any one of the equipments that is available. All dust covers shall be in place on the equipment prior to the initiation of this test. Two technicians shall be used to conduct these tests but only one shall be used per repair action as defined in paragraph 4.6.2. The experience level

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of the technicians shall not be more than the level as specified in paragraph 3.3.3.2.3.3. Before beginning the test, the equipment shall be checked to determine that it is operating normally. Test equipment, tools, spare parts, diagnostic programs, and documentation shall be available for the test.

4.6.2 Repair Actions - A repair action shall be as follows:

(1) Fault Isolation

(2) Repair

(3) Equipment Checkout

4.6.3 Simulated Failures - A total of 25 failures one at a time shall be introduced into the equipment. The failures shall be randomly introduced throughout the equipment. The points of failures shall be approved by the procuring activity prior to initiation of test.

4.6.3.1 Inducing Failures - Single failures shall be induced into the equipment by the government representative by substituting modules that are known to be defective or, where defective modules are not available, simulated failures will be utilized by grounding points at test jacks or on extended modules. All demonstrated failures will consist of a single solid malfunction.

4.6.4 Repair Time Measurements -The time required for the technician to perform each repair action, including the use of technical manuals and support equipment, shall be measured starting at the time the technician is made aware that a malfunction exists and ending at the time the equipment is restored to normal operations. Any delay, such as waiting for parts, obtaining test equipment, or administrative delays shall be excluded.

4.6.5 Test Reporting - Not later than 30 days after the conclusion of the maintainability demonstration the contractor shall submit to the procuring agency a report of the demonstrations. This report shall include all pertinent data and observations, major problem areas and recommendations for corrective action as required. Subsequently, maintainability predictions and maintenance resource requirements data shall be appropriately updated.

4.7 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require

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modification of such procedures, including additional tests, when deemed necessary.

4.8 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.9 Presubmission Testing -No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.10 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General -All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Intended Use - The equipment covered by this specification is intended to convert data represented by three wire 400 cycle per second (cps) electrical synchro signals into 11 bit binary dgital data and to convert 11 bit binary digital data into three wire 400 cps synchro electrical signals.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115± 1.0.V AC as applicable.
These dataare to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.4 Non-repairable Assemblies - As a general rule non-repairable subassemblies of the production equipments should be encapsulated or hermetically sealed. However, for the preproduction items the non-repairable assemblies are not to be so sealed in order to permit repair by the contractor wherever possible. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a MTBF greater than 5000 hours and for many applications this figure must be nearer to 20,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Type Designation - The parenthesis (*) when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature. The complete type number shall be used in nameplates, shipping records and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements.

6.8 Asterisk - In specification revisions and superseding amendments, an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irresponsive of the asterisk notations and relationship to the last previous issue.

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6.9 Items of Equipment - The equipment covered by this specification shall consist of the following item.

ITEM	TYPE DESIGNATION	APPLICABLE PARAGRAPHS
Synchro Converter	(*)	3.5

6.10 Performance Characteristics - The following listed Synchro Converter Set performance characteristics shall be measured to determine that they are in accordance with the related portions of paragraph 3.5 and the associated subparagraphs.

(1) Accuracy - The Synchro Converter Set shall be tested to determine that the accuracy the D/S and S/D modes are in accordance with the specification.

(2) Conversion speed - Each mode shall be tested to determine that it is performed within the specified time.

(3) Power Consumption -Equipment power consumption shall be measured to determine that it is within the specified limit.

(4) D/S Channel Storage - The D/S mode shall be tested to show that the information stored does not have to be updated.

(5) Primary power variations - The equipment shall be tested to determine the limits of input power over which it will satisfactorily operate. These limits must equal or exceed the specification requirements.

(6) Input/Output electrical characteristics - All input and output circuits shall be tested to determine that they perform as required by the appropriate specification.

(7) Input/Output Channels - All channels in the S/D and D/S mode are to be checked for proper operation and that one channel has no effect on any other.

(8) Maintainability - The equipment shall be tested to determine compliance with maintainability requirements as specified herein.

(9) Electrical Interference - The equipment shall be tested to determine compliance with the interference requirements as specified herein.

6.10.1 Training of Government Inspectors - The contractor shall conduct a one week training course for cognizant government inspectors to assure knowledgeable surveillance of the product during the various tests. This course shall be designed to acquaint the attendees with the functional and mechanical

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features of the equipment especially in the areas of diagnostics, selftest, and maintainability. The course shall also detail the operational tests in order to show the application of each test in proving the functional integrity of the equipment.

6.11 Applicable Documents

SPECIFICATIONS

Military

MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-T-5422(ASG)	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-26482	Connector, Electric, Circular, Minature, Quick Disconnect
MIL-S-20708	Synchros, 60 and 400 cycle, General Specification
MIL-M-7793	Meter, Time Totalizing
MIL-D-7018(WEP)	Drawings, For Avionics Equipment; Procurement of
MIL-P-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-C-38300	Connectors, Electrical, Circular, Multicontact High Environment, Quantitative Reliability General Requirements
MIL-R-23094(WEP)	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-T-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

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6.11 Applicable Documents

SPECIFICATIONS continued

Military

MIL-C-81332(WP)	Computer A-NEW

STANDARDS

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-756	Reliability Prediction

WEAPONS SPECIFICATIONS

WS-3099	Maintainability, General Specification for

OTHER PUBLICATIONS

MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment
BUWEPS Instruction 4700.2a	

6.11.1 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the procuring activity or as directed by the contracting officer.

6.12 Criteria for Multiple Source - The criteria of this paragraph shall be used for determining the extent to which the requirement for multiple independent source availablility of microcircuits is satisfied. For interchangeability at the MIC level, multiple source availability exists if the devices from the different sources may be used interchangeably, device by device, in the final equipment without any need for mechanical or electrical changes. For interchangeability at the card level, multiple source availability exists if a card composed of MICS from one source is mechanically and electrically interchangeable with another card composed of MICS from another source. Numerous aspects need be considered to arrive at the independence of sources; these involve production independence, corporate (or firm) and financial independence, and geographical separation. The production independence of one source to another is determined by the extent to which the one source is independent of the other for manufacture; and by the lack of common original suppliers of raw or processed materials. Design, process information, technology, and diffusion masks may be derived from a common source. However, in such a case, each device source shall have a demonstrated capability of originating these items. The

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corporate (or firm) and financial independence of one source of another is determined by the absence of common directors, common officers, or holders of large common financial interests. Geographical separation is simply the distance between the locations where principal manufacture takes place by two sources.

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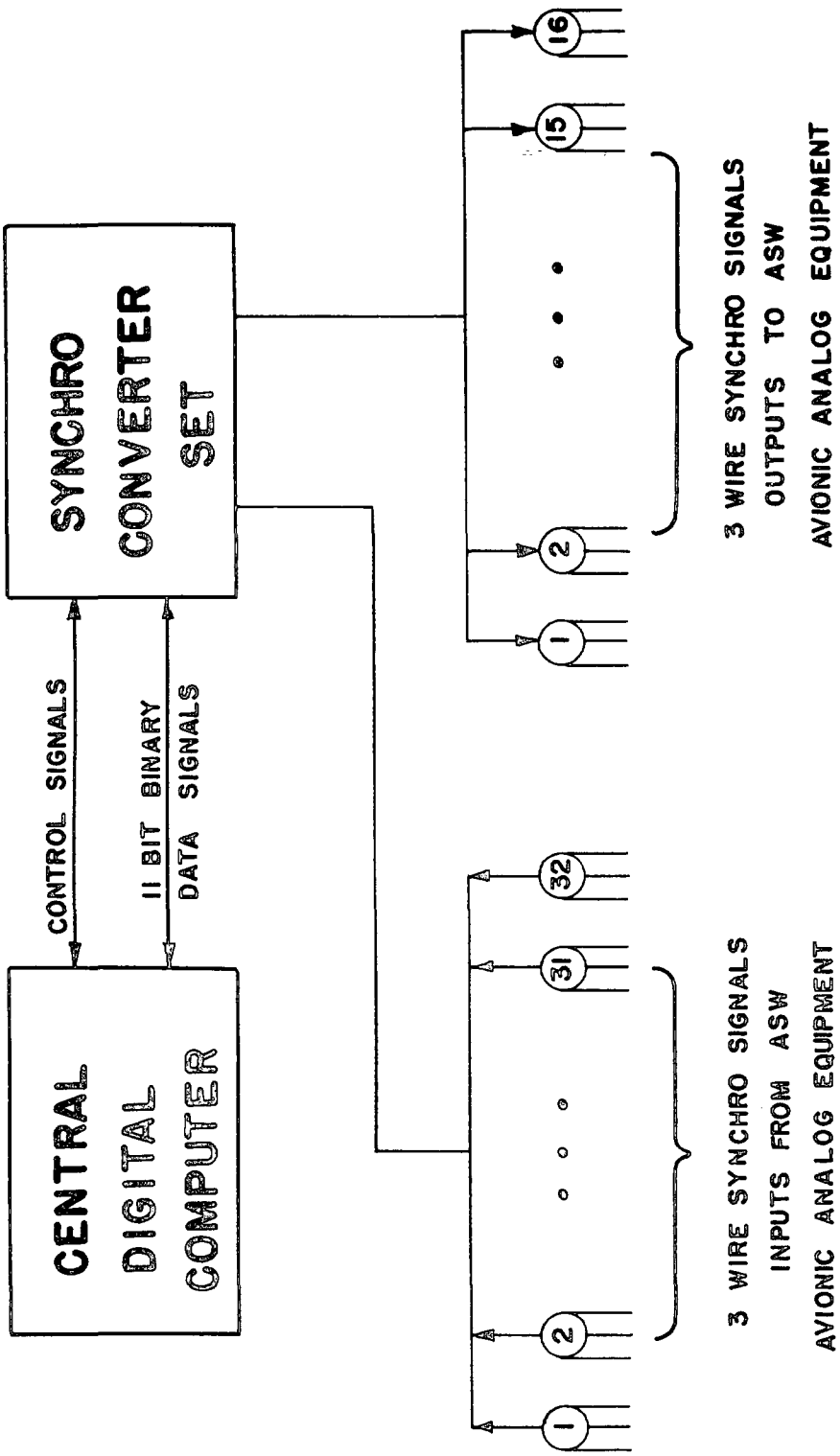


Figure 1 Functional Diagram of Synchro Converter Set

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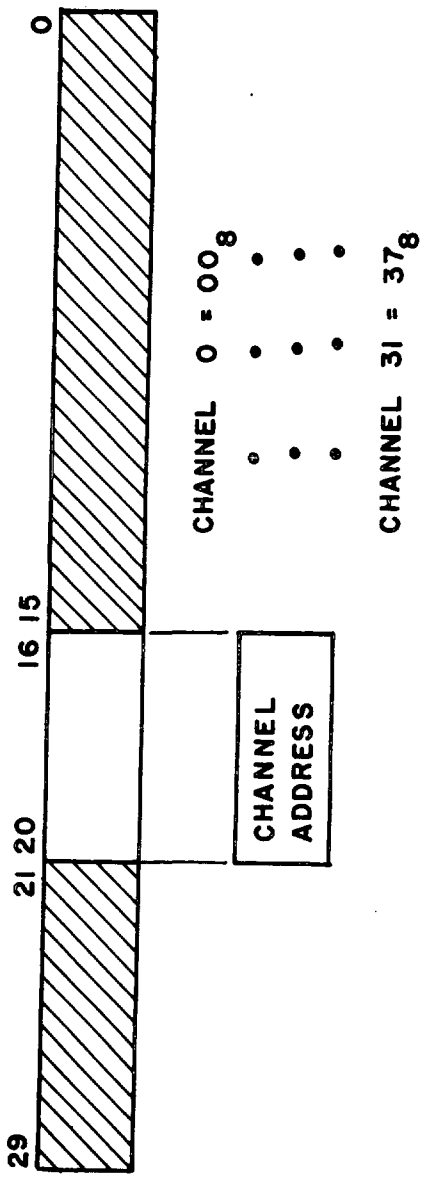


Figure 2 Channel Address Word

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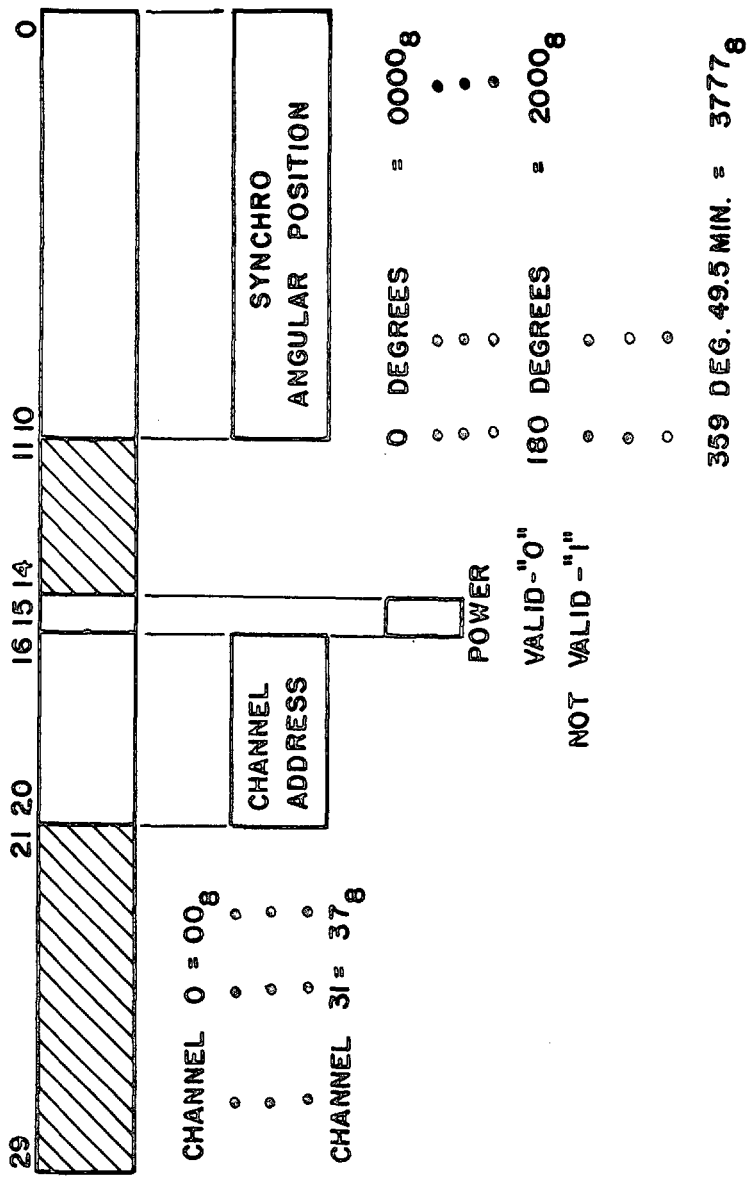


Figure 3 Synchro Angular Position Word

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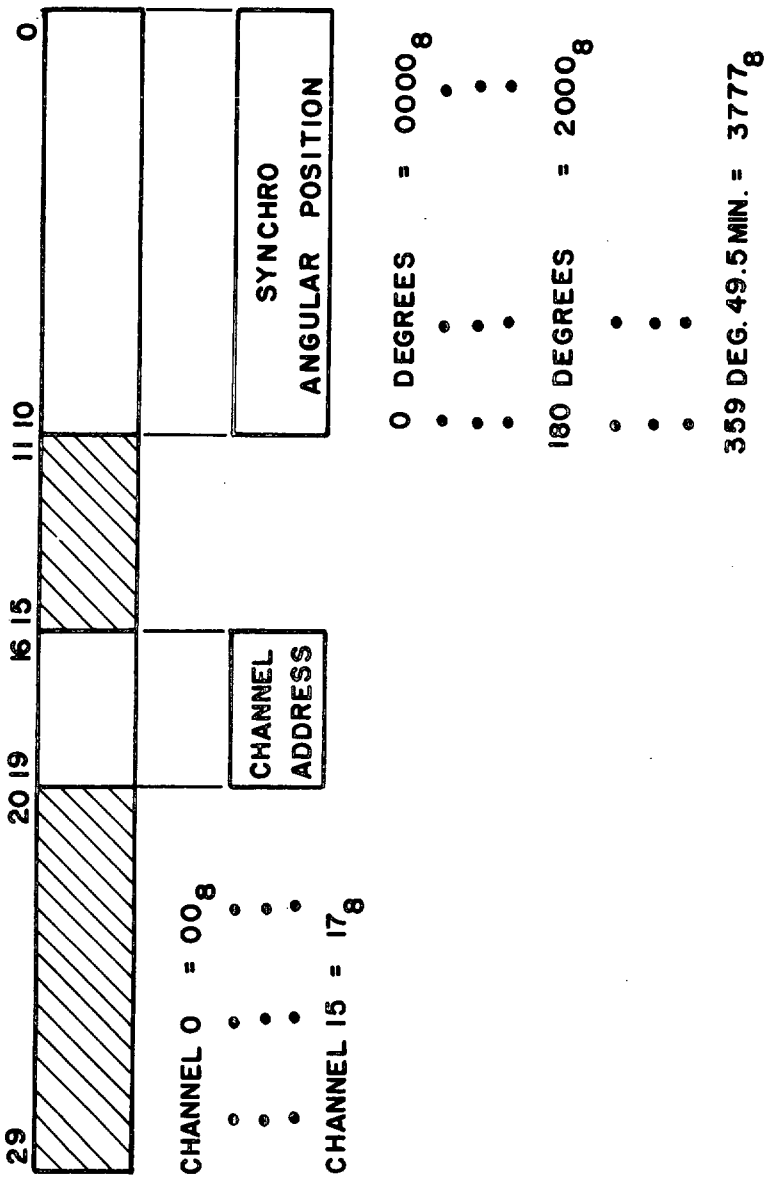


Figure 4 Synchro Angular Position and Channel Address Word

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APPENDIX A

GENERAL SPECIFICATION FOR THIN-FILM INPUT AMPLIFIER

1. SCOPE

This document covers design requirements for thin-film and monolithic digital input amplifiers as described hereunder.

2. APPLICATION

The circuit should provide the equivalent performance of the present discrete input amplifier as shown in Figure 1. This circuit will be used at the termination of 100 feet of twisted pair cable, which is one of four cables being driven by one data line driver as shown in Figure 2.

3. CIRCUIT INPUT CHARACTERISTICS

3.1 Voltage Level Inputs - The binary "one" state shall be 0 volts ±.5 volt and the binary zero shall be 4 volts ±1 volt as measured at the output terminal of the data line driver (See Figure 2).

3.2 Trigger Threshold Level - The threshold level distinguishing the "one" state from the "zero" state is +1.5 volts ±0.5 volts at the input amplifier terminal.

3.3 Noise Rejection - The circuit should be capable of ±5 volts common mode noise rejection. It should be capable of handling noise levels up to ±5 volts peak, without any change in output level.

3.4 Rise and Fall Times - The time required for a data line driver to switch logic levels, as measured from the 10 per cent to 90 per cent of full amplitude points at the output terminals of the driver (before the 100 feet of twisted pair cable), shall not be greater than 1.0 microsecond.

3.5 Loading - The maximum steady state current drawn from a line by an input amplifier circuit shall not exceed 1.5 milliamps.

3.6 Input Impedance - The input amplifier circuit must have an a.c. input impedance of 130 ohms and a d.c. input impedance of 2000 ohms.

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4. CIRCUIT OUTPUT CHARACTERISTICS

4.1 Voltage Level Outputs - The binary "one" state shall be 4 volts ±1 volt and a binary "zero" state shall be 0 volts ±.5 volts.

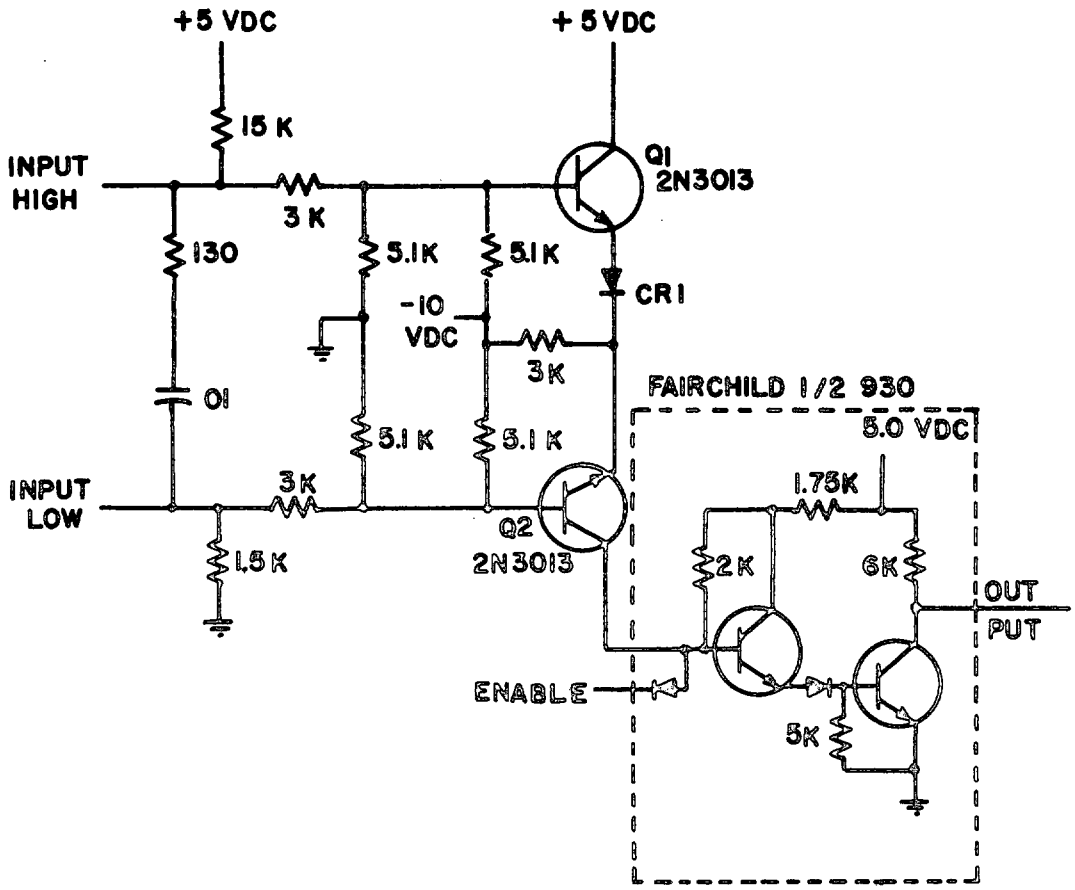
4.2 Rise and Fall Times - The time required for the input amplifier circuit to switch logic levels, as measured from the 10 per cent to 90 per cent of full amplitude points at the output terminals of the amplifier, shall be not greater than 0.5 microsecond when loaded as described below.

4.3 Loading - The circuit must be capable of sinking 30 milliamps of current at the output terminals.

5. TEMPERATURE

The circuits must be capable of meeting all the requirements of this specification over the full military temperature range of -55° C to +125°C.

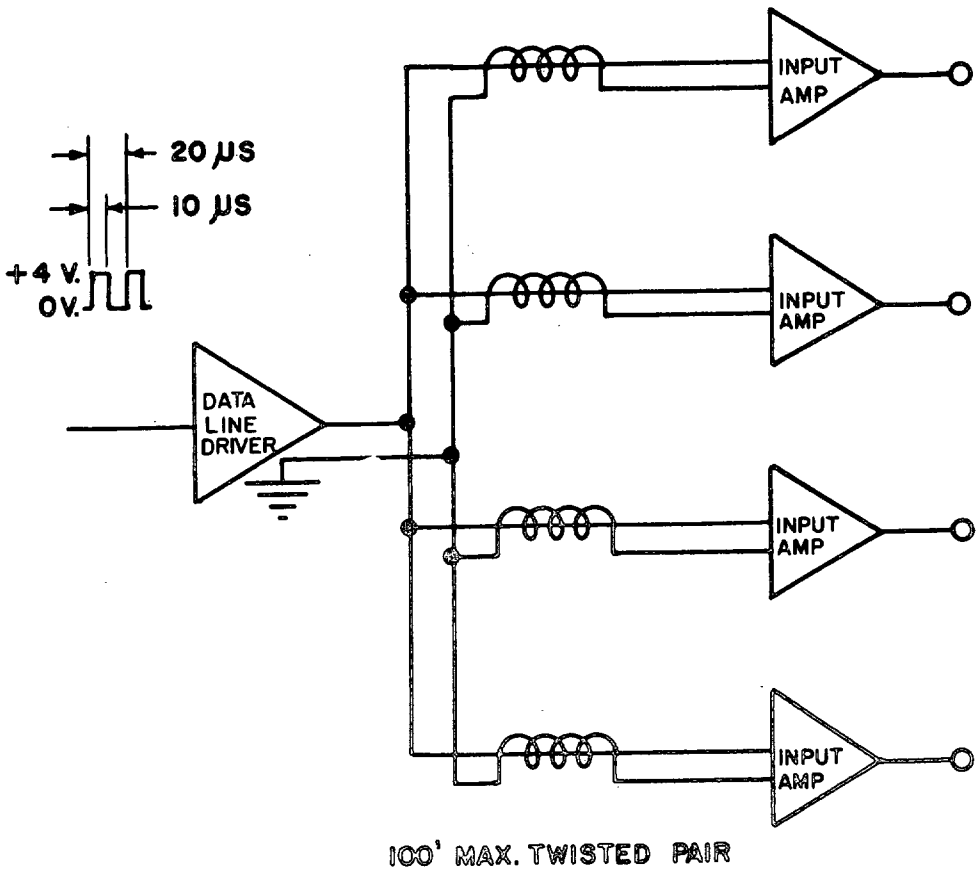
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INPUT AMPLIFIER CIRCUIT

Figure 1

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INPUT/OUTPUT DRIVERS

Figure 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-81344(WP) SYNCHRO CONVERTER SET, A-NEW

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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